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MULTI-OBJECTIVE SIZING FOR HYBRID SOLAR SYSTEMS
USING SYNTHETIC RESIDENTIAL LOADS AND
MACHINE-LEARNING-BASED LONG-TERM
IRRADIANCE-TEMPERATURE FORECASTS

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of the requirements for the award of the degree of Master of Science
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DECLARATION

This research proposal is my original work and to the best of my knowledge it has not been presented for a degree award in this or any other university.

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ABBREVIATIONS

Abbreviation	Full Form
BESS	Battery Energy Storage System
CNN	Convolutional Neural Network
CREST	Centre for Renewable Energy and Sustainable Technologies
CSI	Clear Sky Index
DOD	Depth of Discharge
GHI	Global Horizontal Irradiance
GRU	Gated Recurrent Unit
HRES	Hybrid Renewable Energy System
IRR	Internal Rate of Return
KES	Kenyan Shilling
KPLC	Kenya Power and Lighting Company
LCOE	Levelized Cost of Energy
LPSP	Loss of Power Supply Probability
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MECS	Modern Energy Cooking Services
MECS-CREST	MECS Modified CREST Model
MILP	Mixed Integer Linear Programming
MC	Monte Carlo
NASA POWER	NASA Prediction of Worldwide Energy Resources
NPC	Net Present Cost
NSGA-II	Non-dominated Sorting Genetic Algorithm II
O&M	Operation and Maintenance
PV	Photovoltaic
PVGIS	Photovoltaic Geographical Information System
R^2	Coefficient of Determination
SCR	Self-Consumption Rate
SOC	State of Charge
TMY	Typical Meteorological Year
TOU	Time-of-Use
TUS	Time Use Survey
XGBoost	eXtreme Gradient Boosting

NOMENCLATURE

Symbol	Description
P_{PV}	PV array capacity (kW)
E_{bat}	Battery storage capacity (kWh)
P_{inv}	Hybrid inverter rating (kW)
P_{bat}	Battery power rating (kW)
G_t	Solar irradiance (W/m^2)
T_t	Air temperature ($^{\circ}C$)
L_t	Household load demand (kW)
P_t^{PV}	PV generation at time t (kW)
P_t^c	Battery charging power at time t (kW)
P_t^d	Battery discharging power at time t (kW)
P_t^{grid}	Grid import/export power at time t (kW)
E_t^{ns}	Energy not supplied at time t (kWh)
E_t^{served}	Energy served by PV+battery at time t (kWh)
SOC_t	Battery state of charge at time t
u_t^c	Battery charge switch (0/1)
u_t^d	Battery discharge switch (0/1)
w_t	Representative day weight
c_{PV}	PV capital cost (KES/kW)
c_{bat}	Battery capital cost (KES/kWh)
c_{inv}	Inverter capital cost (KES/kW)
c_{BOS}	Balance of system cost (KES)
G_{STC}	Irradiance at standard test conditions ($1000 W/m^2$)
T_{STC}	Temperature at standard test conditions ($25^{\circ}C$)
γ	PV temperature coefficient ($^{\circ}C^{-1}$)
η_c	Battery charge efficiency
η_d	Battery discharge efficiency
DOD	Depth of discharge
D	Days of autonomy
$\bar{L}_{daily}$	Weighted average daily load (kWh/day)
n_t	Active residents at time t
$p_{i,t}$	Appliance i switch-on probability at time t
$P(n_t n_{t-1})$	Markov occupancy transition probability
$TOU_{i,t}$	Time-of-use probability for appliance i at time t
$p_{light,t}$	Lighting switch-on probability at time t
$irradiance_t$	Irradiance at time t for lighting model
CSI_t	Clear Sky Index at time t
$GHI_{clear-sky,t}$	Clear sky GHI at time t
T_L	Linke turbidity coefficient
h_t	Hidden state at time t (neural networks)

Symbol	Description
$\mathbf{X}_{t-11:t}$	Input window (12 months)
z_t	GRU update gate
$\tilde{\mathbf{h}}_t$	GRU candidate activation
$\hat{y}_m$	Forecasted value for month m
σ_m	Forecast uncertainty for month m
$f_1(x)$	Cost objective function
$f_2(x)$	Self-sufficiency objective function
ϵ_i	ϵ -constraint level i
$\mathcal{P}$	Pareto front
σ_{model}	Model standard deviation
σ_{meas}	Measured standard deviation
$L_{\text{model},t}$	Model load at time t
$L_{\text{meas},t}$	Measured load at time t
Δt	Time step

ABSTRACT

This study proposes a unified framework for the optimal sizing of residential hybrid photovoltaic–battery–grid systems. It is tailored to solar installers working with unmetered households. The work addresses the gap between advanced multi-objective optimization and practical, installer-level design by linking household-resolved load profiling, long-term solar-irradiance and temperature forecasting, and mixed-integer linear programming-based sizing into a single workflow. This should yield three clear design modes: cheapest, balanced, and high-autonomy.

The methodology starts with a stochastic bottom-up load model that generates one-minute weekday and weekend demand profiles from a short online survey on appliances, occupancy, time-of-use patterns, and lighting. Using household-specific Markov occupancy transitions, appliance-use probabilities, and daylight-responsive lighting, the model produces Monte Carlo ensembles that are validated against a 30-day monitored case-study household using mean absolute error, variability ratios, and peak-to-average metrics. In the second stage, a physics-informed deep-learning ensemble is trained on NASA POWER data (1981–2025) to forecast one-year-ahead daily Global Horizontal Irradiance and two-meter air temperature at the study site. Multiple architectures such as convolutional neural network–long short-term memory(CNN-LSTM), convolutional neural network–gated recurrent unit(CNN-GRU), convolutional neural network(CNN), gated recurrent unit(GRU), extreme gradient boosting(XGBoost), and random forest are benchmarked. The best-performing model is selected and equipped with Monte Carlo dropout to provide daily prediction intervals. In the third stage, mixed-integer linear programming in MATLAB optimizes photovoltaic capacity, battery energy, and inverter power across three Pareto-optimal design modes built from percentile-based represen-

tative days: cheapest (median load, high solar), balanced (median load, median solar), and high-autonomy (high load, low solar).

The framework translates minimal survey inputs into explicit, installer-friendly specifications for residential photovoltaic–battery–grid systems. The designs are also grounded in probabilistic load and solar-irradiance scenarios. By integrating household-resolved modelling, long-term, uncertainty-aware solar forecasting, and multi-objective sizing, this work bridges advanced research on hybrid renewable energy systems(HRES) with the practical requirements of residential solar installation.

1 INTRODUCTION

1.1 Background

Factors like geopolitical and economic instabilities threaten the future energy outlook of the world. Energy has also been hit by recent events in the Ukraine and Middle East. Global electricity demand is outstripping supply, and this is resulting in global reliability issues. Increase in urban electrification, electrification of transport, industry and heat, is straining the grid. Air pollution in the energy sector contributes to more than 6 million premature deaths and 90% of the global population live in polluted air [1]. The availability of electricity is not only stagnating but also in certain cases reversing.

Nevertheless, global renewable capacity is set to more than double by 2030. This means adding over 5,500 GW of new capacity worldwide. That is 25% above current government targets and 70% more growth than seen from 2017 to 2023. To hit this, annual additions need to reach 940 GW. [2]. Solar PV and wind will drive 95% of all renewable capacity growth through 2030. Within this solar and wind growth, solar PV alone accounts for 80%. China leads with 60% of the growth, already on track to achieve its 1,200 GW solar and wind goal six years early, through competitive auctions. EU, India and US are also scaling up. The US are promoting solar with Inflation Reduction Act (IRA) tax credits, the EU plans 600 GW of solar in competitive auctions and corporate power purchase agreements (PPAs), and India has the fastest-growing renewable energy market in the world [2]. Due to national policies, renewables are growing to be low-cost power across the globe in over 140 countries.

The solar boom is not just restricted to large scale power plants but is now expanding to homes worldwide. The addition of new manufacturing capacity (1,100 GW a year by 2024) has driven prices down more than 50% since the onset of 2023, making rooftop solar

affordable to millions of households [2]. Investments around the world are over 1 billion per day with customers looking to save on their power bills and to gain energy security. There are however problems with the transition to decentralised generation. Over 1,650 GW (an increase of 150 GW over 2022) of solar projects are now awaiting to connect to the grid, showing the challenges of system integration [1,2]. This shows the necessity of back-up technologies to support solar, including energy storage and intelligent energy management.

Solar systems require complementary technologies like batteries explaining the increasing number of residential hybrid renewable energy systems (HRES) modernities. HRES integrate rooftop PV, batteries and grid connections into a single system to ensure power during blackouts, peak electricity rates or overcast days [3]. Residential deployment is becoming increasingly popular all over the world as individuals aim at becoming self-sufficient and insuring against the increasing electricity prices. Lithium-iron-phosphate batteries are favored due to their safety and durability and are combined with hybrid inverters to achieve efficiency in managing energy in different applications [4].

Residential HRES still face technical challenges that hinder adoption. Sizing the system involves matching PV output, battery storage and dynamic loads, with factors such as weather uncertainty and grid conditions [3]. Integration issues include managing charge and discharge, avoiding over-voltage and incompatibilities between components. These challenges require advanced control schemes, which many installers are unable to carry out properly [4]. Some homes also still lack the necessary infrastructure such as finances to support solar adoption.

Favourable market dynamics, dropping battery prices, plentiful solar capacity and high electricity prices make HRES financially attractive across the globe. And policy

incentives and corporate sustainability goals boost demand, making residential HRES a key element in the future of distributed energy systems [3, 4]. This presents an opportunity to develop design approaches to enable installers to deliver efficient systems at scale.

1.2 Problem statement

Residential solar installers face three design constraints that are critical in undermining the adoption of the systems. First, installers do not have access to the residential household load profile as the some of the installation sites do not have power meters, and clients usually do not accept weeks-long monitoring campaigns to obtain the accurate demand profile [5]. Second, designers use the historical average sun-hours for PV output design and do not consider the future variation of irradiance between seasons, weather conditions, and climate [5]. Third, the current sizing techniques are project-based, maximizing lifecycle economic indicators to justify projects [6], unlike installer-based objectives that balance between initial capital investment and grid independence to influence client purchasing choices.

In the absence of load profiles, installers set the PV-battery capacity that mismatches household consumption, resulting in an oversized system that wastes money or an undersized system that provides insufficient power to the household [7]. Historical sun-hour averages overlook variations in PV production in the future, leading to continuous underperformance of energy yield. Finally, the lack of clear, installer-friendly solar system options leaves clients confused about what systems to purchase.

Without intervention, customers continue getting subpar investment value while installers lose credibility due to failed systems where design flaws manifest [7]. This results in dissatisfaction which spreads among people and negatively affects solar technology

and market growth. Load estimation, variability aware PV sizing, and client centric system design are essential tools that need to be given to the installers.

1.3 Objectives

The main objective of the research is to develop a complete framework that combines the residential load modelling, long term prediction of solar irradiance and temperature, and multi-objective sizing of hybrid solar system.

To achieve the main objective, the following specific objectives will be accomplished:

1. To develop a model for generating realistic synthetic residential electricity load profiles.
2. To develop a machine learning model for predicting long-term solar irradiance and temperature.
3. To develop a multi-objective optimization approach to sizing hybrid solar systems.

1.4 Justification and Scope of Study

The study justifies an input-process-output approach to solar system design. The framework translates details from the home, such as the list of appliances, the probability of their use, and the location of the home, into useful system designs. The processes used are: synthetic load profiling, machine learning for long term irradiance and temperature forecasting, and multi-objective optimization to minimize cost and maximize grid independence. There are 3 custom design packages offered by the processes: lowest cost, balanced, and maximum grid independence. The three packages include PV array size, battery capacity, inveter sizing and metrics. Weeks-long monitoring is replaced with instant synthetic load profiles. Solar output prediction provides good yield predictions.

The optimization offers good trade-offs between cost and autonomy. Installers receive faster evaluations, greater close rates via clear-cut choices, and reduced system issues as a result of better-sized systems. Access to affordable and reliable power is ensured, in accordance with their budget constraints and independence.

The study scope is limited to residential hybrid PV-battery-grid systems. This does not include commercial systems, industrial systems and other generation sources. The intended audience is solar installers and solar system designers. The technical scope includes the standard domestic appliances, PV panels, LiFePO₄ batteries, hybrid inverters and flat residential electricity tariffs. Data scope covers household surveys, NASA POWER dataset of solar data, and one-month field measurement to validate the model.

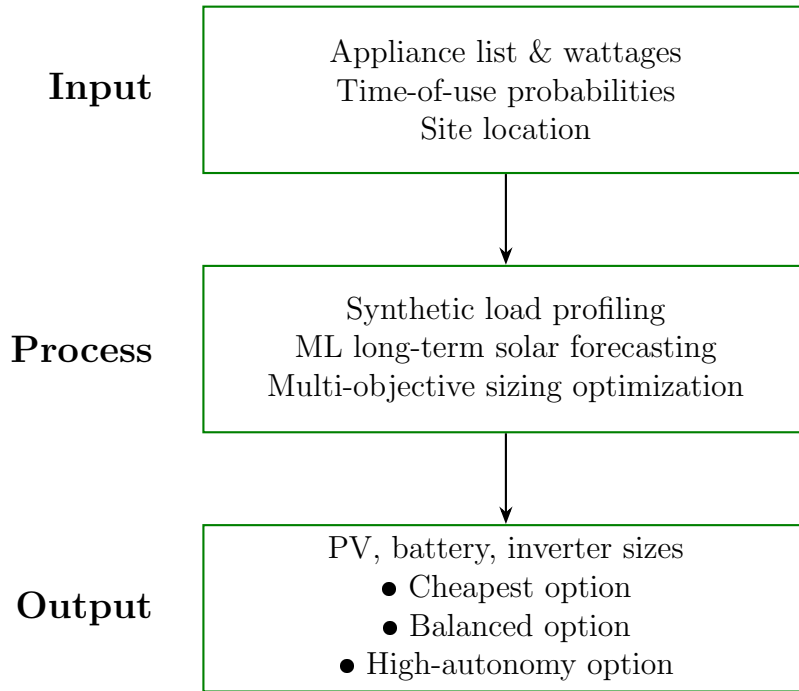


Figure 1: Input-Process-Output Framework for Residential Solar System Design.

2 LITERATURE REVIEW

2.1 Residential Solar system design

The design of a residential solar system is the process of choosing and configuring the key components of a home-scale solar energy system in a manner that the components collectively meet the electricity needs of the household without violating technical, economic, and reliability requirements [3]. The aim is to identify the right sizes of the PV generator, battery bank and inverter so that the system could supply the load of the home safely, reliably and at a reasonable cost over its lifetime, under the local climatic conditions, grid conditions and user preferences [8].

2.1.1 Core components of a residential solar system

An average residential solar system is constructed on a few main building blocks which collaborate to transform, store and supply electricity to the home. The photovoltaic array is a set of solar panels which change the sunlight into direct current (DC) electricity. The size of the array is typically measured in the unit kilowattspeak (kWp) and represents the amount of power that the array can generate under typical test conditions. Battery storage comprises one or more rechargeable batteries, typically of lithium-based materials such as LiFePO_4 , which store excess solar energy for use when the sun is not shining. The battery can be characterised by its energy capacity (kWh), power rating (kW), and the behaviour of its operational limits, like state-of-charge (SOC) ranges and cycle-life behaviour. The power-electronic device that converts DC power provided by the PV array and battery into alternating current (AC) power to power household appliances is known as the inverter, or in more advanced systems, the hybrid inverter. In a hybrid configuration, the inverter also manages the flow of power between PV, battery, load, and

the grid, allowing it to charge the battery from either PV or grid and to export surplus energy when permitted [9]. In many installations, the dwelling remains connected to the utility grid, which can supply power whenever solar generation and stored energy are insufficient and can receive any surplus electricity through net-metering or feed-in arrangements [2]. The combination of these elements and connection to the grid or off-grid determines whether the system will be categorised as grid-tied, off-grid, or a hybrid (grid-tied with battery storage).

2.1.2 Grid-tied systems

A grid-tied solar system is directly connected to the utility grid, without a battery in the system to store energy. The household load is powered by the solar power generated during the day, and any excess production is fed back to the grid. When the PV output is insufficient for example at night or during cloudy periods, the grid supplies the missing power. Grid-tied systems are typically the simplest and least expensive configuration, as they avoid the cost of batteries while still reducing electricity bills and grid demand [3]. Most of them, however, are typically unable to supply power in the event of a grid outage unless they are provided with extra safety equipment or are islanded.

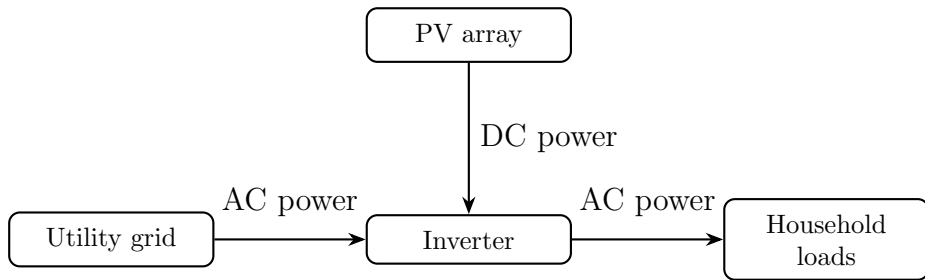


Figure 2: Grid-tied residential PV system (no battery storage).

2.1.3 Off-grid systems

An off-grid or stand-alone solar system operates independently of the utility grid and is often used in remote or rural locations where grid connection is unavailable or prohibitively expensive. The PV array charges the battery bank, and the inverter supplies AC power to the household load from the battery [5]. Because there is no grid backup, the system must be sized to cover the load even during periods of low solar irradiance, which generally requires larger PV arrays and higher battery capacity to ensure security of supply. Off-grid systems are thus more expensive and challenging to design, but they give full energy independence from the grid [8].

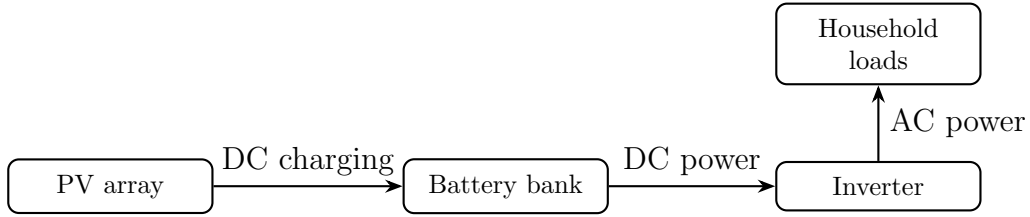


Figure 3: Off-grid residential PV–battery system.

2.1.4 Hybrid (grid-tied with battery storage) systems

A hybrid PV–battery–grid system combines elements of both grid-tied and off-grid configurations. The dwelling remains connected to the grid, and a battery bank is installed alongside the PV array. Under normal conditions, solar energy supplies the household load first, with remaining surplus either charging the battery or being exported to the grid. During the evening or grid outages, the battery can discharge to cover the load, and the grid can still supply additional power when needed. This configuration offers flexibility. It can reduce electricity bills like a simple grid-tied system, provide backup power during outages, and better manage peak demand on the grid [6]. Hybrid systems

thus become a popular option in those regions where there are frequent blackouts or time-of-use tariffs and where the cost of stored energy is high.

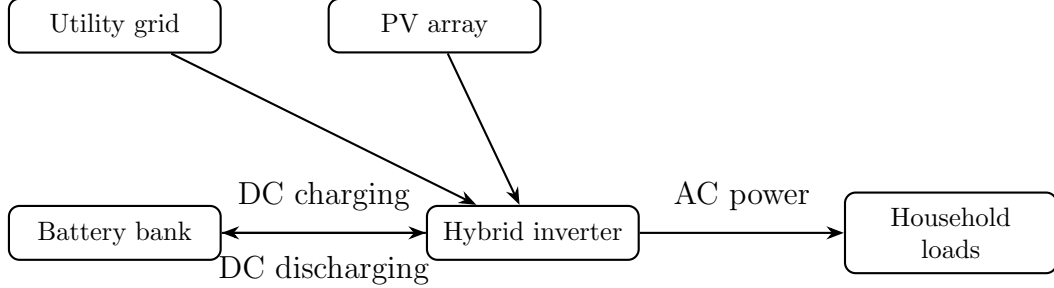


Figure 4: Hybrid grid-tied PV–battery–grid residential system.

In each of the three configurations, the design problem remains essentially the same, namely, to select PV capacity, battery capacity, and the size of an inverter such that the system will meet the demand and reliability requirements of the household at the minimum cost, without violating the physical and operational constraints. [8].

2.2 Residential load profiling

The process of creating time-series representations of household electricity demand, typically at sub-hourly or minute resolutions, such that these profiles can be used to design systems, do demand-response planning, or study grid-impact effects, is known as residential load profiling [10]. The fundamental issue is to model the strong intra-hour variability, seasonal changes and household-specific patterns of behaviour which occur due to staggered appliance use, seasonal variations and household-specific patterns of behaviour. In the last couple of years, two generalised families of techniques have arisen: physically measured load curves and synthetic or model-based load profiles.

2.2.1 Load metering and physical measurement

Electrical load metering, with high-resolution meters like power loggers or smart meters, is the most direct method of obtaining residential load profiles [7]. They capture aggregated or disaggregated power or energy usage at time steps of seconds to minutes, depending on the hardware and data-logging configuration. Normal deployments produce days to weeks of continuous measurements, which may be further used to capture the daily, weekly, and seasonal trends, and the rare occurrences, such as the spikes of high-power appliances or the period of low use.

Load profiles from field measurements are widely considered as the gold standard since they reflect the actual behaviour of particular households, such as peculiar appliance loads and occupancy patterns [11]. They are particularly helpful when validating model-generated synthetic profiles, as well as in calibration of statistical quantities such as appliance-use probabilities and duration distributions. But load metering is also resource-intensive, as it requires access to the site, client permission, a safety-compliant installation, and a long-term data-management infrastructure. In low-engagement or unmetered households, this technique might be impractical, and the use of model-based load-profile generation techniques based on surveys and behavioural statistics instead of continuous meter data might be considered.

2.2.2 Bottom-up vs top-down load modelling

In addition to physical measurement, researchers make a broad distinction between bottom-up and top-down load-modelling methods. In a top-down model, aggregate statistics such as annual or monthly energy use, average load factors, or peak-to-average ratios are used to reconstruct stylised load curves of representative consumer groups or

the whole residential sector [10]. The benefit of top-down approaches is that they are scalable since they can swiftly produce national- or city-scale load profiles at a relatively low cost using energy statistics and weather data that are widely available.

In contrast, bottom-up models begin with the best level of detail: individual appliances, occupants, and activities, and then proceed upwards to rebuild the total household load. These models explicitly describe the appliance inventory in a dwelling, whether and how often the appliances would be used and the effect of occupancy and external conditions, including the use of lights in a home during low daylight [10]. The bottom-up approaches are thus intrinsically well-suited to the issue of capturing the heterogeneity of households, as well as exploring scenarios such as the introduction of new appliances, solar PV, or demand-response programmes. Nevertheless, they demand fine-grained input data, and hence, there is a trade-off between the fidelity of the model and its availability. Hybrid strategies that involve bottom-up disaggregation and top-down calibration are becoming the norm, as they take advantage of the strengths of both paradigms.

2.2.3 Stochastic household load models

Stochastic household load models have emerged as a key instrument in the generation of high-resolution synthetic load profiles which capture realistic variability. These models model load as a probabilistic process, driven by occupancy behaviour, e.g. transitions between presence and absence states are often modelled via Markov-chain processes, appliance-use behaviour such as randomised on-times, durations and power levels are often modelled by distributions, and by environmental influences such as lighting use modulated by daylight availability or space-heating and cooling driven by temperature. A recent and high-profile example is the CREST family of models, which, starting with time-use surveys and appliance-ownership statistics, builds up stochastic, one-minute-

Table 1: Proposed model vs key predecessors in stochastic household load modelling.

Aspect	CREST 1.0B [10]	MECS-CREST [12]	Proposed model
Resolution	1-minute	1-minute	1-minute
Household Rep.	Random synthetic household details per run	Fixed baseline household details per user story	Fixed baseline household details per household
No. of Residents	User defined (1–6)	User defined (cooking diary based)	Survey-specified fixed number
Day	Weekday & Weekend	Single day	Weekday & weekend day $\times$ MC diversity $\times$ seasonal scaling
Clearness	Loughborough clearness stats for lighting	PVGIS clearness stats for lighting	NASA POWER clearness stats for lighting
Data Sources	UK TUS/ownership stats for average household	Kenyan Cooking diaries + simplified TUS for average household	Specific household TUS/ownership data from surveys tailored for household accuracy
Development Flow	Calibrated to national aggregates	Cooking diaries drive parameters	Household surveys drive parameters
Occupancy	Markov transitions from (UK tables)	Fixed occupancy profile built from cooking diaries	Markov transitions from (Household data)
Appliance Handling	Random ownership/power distributions	User-specified + cooking appliances	Survey-specified ownership
Appliance model	TUS activity probabilities	Simplified TOU + cooking	Survey TUS activity probabilities
Lighting model	Irradiance threshold + random bulbs	Specified LED configurations + irradiance	Survey bulbs + NASA irradiance
Validation Data	Aggregate UK statistics	Illustrative examples only	Independent case-study measurements
Diversity/Scale	Aggregate many runs	Change parameters within/across households	Monte Carlo ensembles
Implementation	Excel/VBA spreadsheet	Modified Excel	Python
Validation Metrics	Calibrated to UK aggregates	Visual examples (Kenya diaries)	MAE, var, P/A ratio vs measured case-study
Output	Single 24h load profile	24h household + aggregate profiles	Weekday + weekend statistical load profiles

resolution load profiles of individual households or settlements. [10].

Ensembles of load curves are naturally produced by stochastic models, and it is often important to sample a variety of realisations to capture the day-to-day variability in when and how long appliances are used. This ensemble picture is especially useful in the scaling of hybrid PV-battery systems-grids since it provides the exposure of tail behaviours such as unusually large or small-load days, which deterministic averages might obscure. More recent work extends these concepts to Markov-based occupant-behaviour simulators and large-scale bottom-up generators capable of synthesising load profiles of thousands of households at once, whilst maintaining household-level granularity to energy-system design studies. [10, 12].

2.3 Solar irradiance and temperature forecasting

Solar irradiance and temperature forecasting plays a central role in the design and operation of residential PV–battery–grid systems, since the expected solar generation directly determines the required size of the PV array and battery bank, as well as the reliability of grid independence [13]. Proper forecasts over a period of minutes to years will enable the system designers to consider the intra-day variability, seasonal changes and long-term climate patterns rather than only relying on the deterministic average. In the last decade, three general classes of approaches have appeared: statistical and classical machine learning techniques, machine learning techniques based on deep learning, and physics-informed or hybrid forecasting models that incorporate numerical weather models with data-driven components.

An important facilitator of these approaches is the accessibility of long-term and high-quality irradiance and temperature data, such as NASA POWER, which provides high-quality, satellite-derived time-series spanning multiple decades and a wide range

of climatic zones [13, 14]. These datasets are especially well-suited to the development of solar-forecasting models because they provide consistent, gridded irradiance and meteorological variables at a time resolution of less than a day, which is suitable for the development of both short-term and long-term forecasting models. In this work, the NASA POWER is selected as the main source of data as it is globally available and extensively validated against ground measurements.

2.3.1 Statistical and machine learning-based forecasting

The oldest and most basic solar forecasting models are based on statistical models that take advantage of historical correlations between irradiance, temperature, and time-dependent features such as hour of day, day of year, or moving-average values of previous solar generation [13]. Linear regression and autoregressive integrated moving average (ARIMA) models, with their variants, are capable of capturing seasonal effects and mean behaviour whilst being computationally lightweight and interpretable. These models are often used for short-term or mid-term horizons where trends and periodicity dominate over abrupt weather changes.

More recently, researchers have extended this line of work to classical machine learning (ML) methods such as decision trees, random forests, and extreme gradient boosting (XGBoost) [13, 15]. These algorithms can handle non-linear relationships and interactions between multiple input variables, including irradiance, temperature, humidity, pressure, and calendar features, and they are particularly effective when the training dataset is of moderate size and clean. ML-based solar and temperature models are widely used in PV system planning and grid-integration studies because they require relatively little domain-specific tuning, yet they can outperform purely linear statistical models in many practical settings [14].

2.3.2 Deep-learning-based forecasting

As high-resolution solar and meteorological data have become more widely available, deep-learning-based approaches have emerged as a dominant paradigm for solar irradiance and temperature forecasting. Such models typically work with time series data and learn complex time series patterns directly out of sequences of historic measurements [13]. The network extrapolates internal representations of the raw or slightly processed data encoding intraday cycles, day-to-day persistence and seasonal changes.

Recurrent neural networks (RNNs), and in particular long short term memory (LSTM) and gated recurrent unit (GRU) networks, have been widely applied to multi-step solar-forecasting problems because they can capture long-term dependencies across days and months [13]. In an LSTM, the hidden state at each time step is updated through three internal gates (forget, input, and output) that decide which information to retain, which to update, and which to expose, allowing the model to selectively remember or forget relevant patterns over long horizons. This is especially useful for solar irradiance, where morning and afternoon ramp up and ramp down behaviours, cloud passing events, and day to day persistence must all be encoded without manually defining separate rules for each regime. GRU networks can provide a similar effect with a slightly simpler gate structure, which can often be found to provide comparable performance at a lower computational cost.

These capabilities are further expanded by convolutional neural networks (CNNs) and their hybrid equivalents, such as CNN-LSTM and CNN-GRU, which extract local spatio-temporal patterns in the input data and then pass them through recurrent layers [13]. In a CNN-LSTM architecture, one-dimensional convolutional kernels slide over the time axis of the input sequence, detecting local shapes such as daily sunrise slopes,

peaks, and short-duration cloud effects; these compressed feature maps are then fed into an LSTM or GRU to model the longer-term temporal evolution. This combination enables the model to exploit both the fine-grained structure of the daily PV profile and the slower-varying seasonal and climatic trends.

Deep-learning-based models have demonstrated state of the art performance on global solar-forecasting benchmarks, especially when trained on long-term satellite-based or re-analysis datasets such as NASA POWER [13]. By learning from multi-year sequences of high-resolution irradiance and temperature data, these models can implicitly capture the statistical structure of weather regimes, including typical cloud patterns, persistence of clear-sky conditions, and inter-annual variability linked to large-scale climate modes. When properly regularized and validated, CNN-LSTM and similar architectures provide accurate, robust forecasts that can be directly integrated into probabilistic or scenario-based design workflows for hybrid PV-battery-grid systems.

2.3.3 Physics-informed and hybrid forecasting

Whereas deep-learning models can be very effective at fitting complex patterns to historical data, they may be challenged when attempting to extrapolate to unseen climates or when the training data available is sparse. To solve this, scientists have created physics-informed and hybrid forecasting models that integrate data-driven techniques with the physical or semi-empirical relationships that are based on the theory of solar radiation and meteorological processes. [14]. A common strategy is to augment the model inputs with a clear-sky index or other physically meaningful features that encode the expected maximum irradiance under ideal atmospheric conditions, allowing the data-driven component to learn only the residuals relative to a physically defensible baseline [13].

In the context of PV system design, physics-informed approaches are particularly

valuable because they provide better extrapolation behaviour and more interpretable errors. They can distinguish between uncertainty arising from imperfect weather models and uncertainty arising from noisy observations. Hybrid models that combine numerical weather prediction outputs with deep-learning components or classical ML regressors have become increasingly popular, as they leverage the strengths of both physics and data while reducing over-fitting to local patterns. Models that use the output of numerical weather prediction as input and deep-learned components or classical ML regressors have become particularly popular, as they leverage the strengths of both physics and data and reduce over-fitting to local patterns. [14].

2.4 Sizing and optimization of PV–battery–grid systems

Sizing and optimization of hybrid PV-battery-grid is aimed at identifying the optimal capacities of the PV array, battery storage and related power-conversion hardware so that the system can meet the reliability and autonomy specifications and remain economically viable throughout its lifetime [9]. Recent research has not focused on rule-of-thumb rules but the formal optimisation models explicitly balancing the cost of investments, self-sufficiency, and grid-dependability, often under multi-year or probabilistic weather and load condition [8].

2.4.1 Optimal sizing of hybrid PV–battery systems

Optimal sizing of hybrid PV–battery systems typically involves deciding the number and layout of PV modules, the usable energy and power capacity of the battery bank, and the inverter rating, subject to constraints such as maximum daily load, available roof area, and technical limits on battery depth-of-discharge and inverter overload. Single-objective approaches often minimize a composite cost metric such as the levelized cost of electricity

(LCOE) or total annualized cost, implicitly treating reliability and self-sufficiency as hard constraints [16]. These techniques are computationally efficient and can be applied to initial design, but can mask trade-offs between low initial cost and high grid-independence.

In contrast, multi-objective size frameworks consider cost, self-reliance (fraction of load covered by PV and battery), and reliability (e.g., loss of power supply probability or unserved energy) as independent goals to be optimised jointly. [9]. This can enable the designer to observe how augmenting PV and battery capacities will decrease grid dependence at a higher capital cost, as well as to locate knee points in the trade-off space. These models can also capture the effect of tariff structure, peak-valley pricing and grid-availability constraints on the optimal sizing of each piece in the PV-battery-grid nexus [15].

2.4.2 Treatment of grid, PV, battery, and inverter in sizing models

The model of the grid in modern sizing formulations assumes the grid can be characterised as an external power source and sink with time-varying tariffs and availability constraints, and the PV array is represented in terms of its peak power rating and its time-series generation profile based on solar-irradiance forecasts [13]. The battery is characterised by its energy capacity, power rating, round-trip efficiency, and state-of-charge limits, and its operation is controlled by charge and discharge constraints that prevent overcurrent and over-discharge. Inverter models capture bidirectional power flow between PV battery, loads, and grid, often including limits of continuous and short-term overload and ensuring that PV or battery power is not converted past the rated power of the inverter [3].

These components are connected by an energy-balance equation at every time step, which requires the sum of PV generation, battery discharge and grid import (when applicable) equals the sum of load demand, battery charge and grid export (when applica-

ble). Grid-interaction decisions, such as selling excess PV production or importing during low-solar-irradiance periods, are then determined endogenously by the optimization, subject to tariff structures and contractual constraints [9]. This integrated representation enables the sizing model to explore the physical feasibility of candidate configurations and also the economic and grid-support performance of such candidate configurations under a wide variety of weather and demand realisations.

2.4.3 Optimization techniques for energy systems

Problems of sizing and operational scheduling of PV battery grid systems are most commonly formulated as mixed-integer linear programming (MILP) or mixed-integer programming (MIP) problems, particularly when equipment sizing and discrete inverter or module decisions are combined with continuous power flows [16]. MILP-based models can optimise the sizes of equipment and operation over multi-period horizons whilst considering technical and tariff-based constraints, and enjoy the benefits of global-optimality guarantees and mature solvers, such as Gurobi. Such models are very suitable for deterministic or scenario-based cases where the feasible region is convex, and the number of discrete variables is moderate [17].

More common methods, including genetic algorithms (GA) and particle-swarm optimisation (PSO), are also frequently used, especially when the objective functions become highly non-linear, non-convex or are evaluated inside of simulation tools that can not easily be expressed as a closed-form mathematical program [11]. These algorithms explore the design space through stochastic search and do not require explicit gradient information, making them flexible for complex multi-objective problems and custom constraints. However, they generally provide no guarantees of global optimality, can be sensitive to parameter tuning, and may require many iterations before convergence. In

practice, the Pareto front performance of the two methods, MILP and metaheuristics, is similar, as demonstrated in comparative studies. While MILP tends to be more robust and transparent, metaheuristics work best when the model has a structure too complex or black-box to be formulated as a MILP [3].

2.4.4 Multi-objective and Pareto-based design

For PV–battery–grid systems, multi-objective optimization is typically implemented via methods such as the ε -constraint approach, linear weighting, and evolutionary multi-objective algorithms (e.g., NSGA-IIgi) [11]. In the ε -constraint method, the designer fixes acceptable levels for some objectives (e.g., maximum allowable cost or minimum required self-sufficiency) and then optimizes a single primary objective, such as reliability or grid-dependence, within those bounds. Linear weighting transforms the vector-valued problem into a scalar one by combining objectives into a single weighted sum, at the cost of imposing a priori preferences on the relative importance of cost, autonomy, and reliability [9].

Evolutionary multi-objective frameworks avoid explicit weighting by evolving a population of candidate solutions and using Pareto-dominance rules to approximate the entire trade-off frontier between conflicting objectives [11]. These modes correspond to different operating regimes of the same underlying PV–battery–grid topology and help stakeholders navigate the design space without committing prematurely to a single scalar objective.

Pirouz et al. [6] systematically describe optimization objectives in more than 40 hybrid renewable energy system (HRES) studies, classifying them into economic, technical, and performance-oriented categories. Table 2 provides this objective taxonomy, illustrating common goals such as total cost, peak-shaving, reliability, emissions, market profits,

and battery-storage degradation.

Table 2: HRES optimization objectives taxonomy (from Pirouz et al. 2024)

Category	Objective	Description
Economic	F1: Total Cost	Capital + O&M + Energy Not Supplied penalties (85% of surveyed studies).
	F2: Load Variance	Peak shaving and load-shape flattening to reduce grid stress.
	F3: IRR	Internal Rate of Return used to assess financial viability.
Technical	F4: LPSP	Loss of Power Supply Probability minimization for reliability.
	F5: Emissions	CO ₂ reduction and environmental impact mitigation.
Performance	F6: Market Profits	Energy arbitrage and market participation for revenue maximization.
	F7: BESS Lifespan	Cycle life and degradation optimization to extend battery life.
	F8: Self-Consumption	PV utilization rate enhancement through on-site use and reduced export.

This structured view helps clarify how the objectives incorporated in a multi-objective PV–battery–grid model align with the broader literature [6, 9].

2.5 Identified research gaps

From current literature on hybrid renewable energy systems (HRES), four important gaps are identified motivating the present work.

Gap 1: Limited availability of household load profiles. Most of the literature on load profiling relies on the measurement of smart meters or intensive field campaigns which are impractical to achieve in unmetered homes or where clients do not want to have their home monitored for long periods of time [7, 11]. Demand can be synthesized from appliance and time use data, as done in the bottom-up and stochastic models such as CREST [10, 12], but few studies translate these profiles into practical, household-specific

PV–battery sizing workflows. .

Gap 2: Over-reliance on historical solar-irradiance averages. A large number of sizing and optimization studies continue to use long-term averages or long-term solar resource data which do not explicitly account for uncertainty in the future or variability from year to year [6]. While deep-learning-based forecasting can give a long-term prediction of global solar-irradiance and global temperature [13], validation across diverse regions, comparison with classical baselines, and explicit prediction intervals remain limited. This limits the deployment of strong, uncertainty-aware solar inputs for risk informed sizing.

Gap 3: Stronger focus on abstract project-level objectives than installer-level options. Multi-objective sizing studies typically optimize life-cycle cost, IRR, or LPSP, which are more relevant to financiers than to installers or end-clients [6, 9, 11]. Only a small number of frameworks explicitly translate the Pareto front into three clear, interpretable design modes—cheapest, balanced, and high-autonomy—that can be directly presented to clients in terms of upfront capital, grid-dependence, and PV–battery–inverter capacities. This partial disconnect hinders the use of advanced optimization in on-ground design practice.

Gap 4: Limited integration of sizing components into a unified workflow. Strong individual components have emerged in the literature (household load profiling, solar-irradiance and temperature forecasting, and sizing using MILP or metaheuristics) that are often applied separately to the problem [3, 9, 10, 13]. Only a few frameworks combine survey-based stochastic load generation, AI-based solar-resource forecasting and multi-objective MILP optimization into a single end-to-end design workflow for residential PV–battery–grid systems. This division restricts the usefulness of research findings

for putting them into practice in an installer-friendly product.

The aim of the present work is to fill these gaps by integrating stochastic load profiles, deep-learning-based solar-irradiance and temperature forecasting with multi-objective MILP optimization to produce three clear design modes (cheapest, balanced, high-autonomy), in which PV, battery and inverter specifications are explicitly defined for specific households and grid conditions.

3 METHODOLOGY

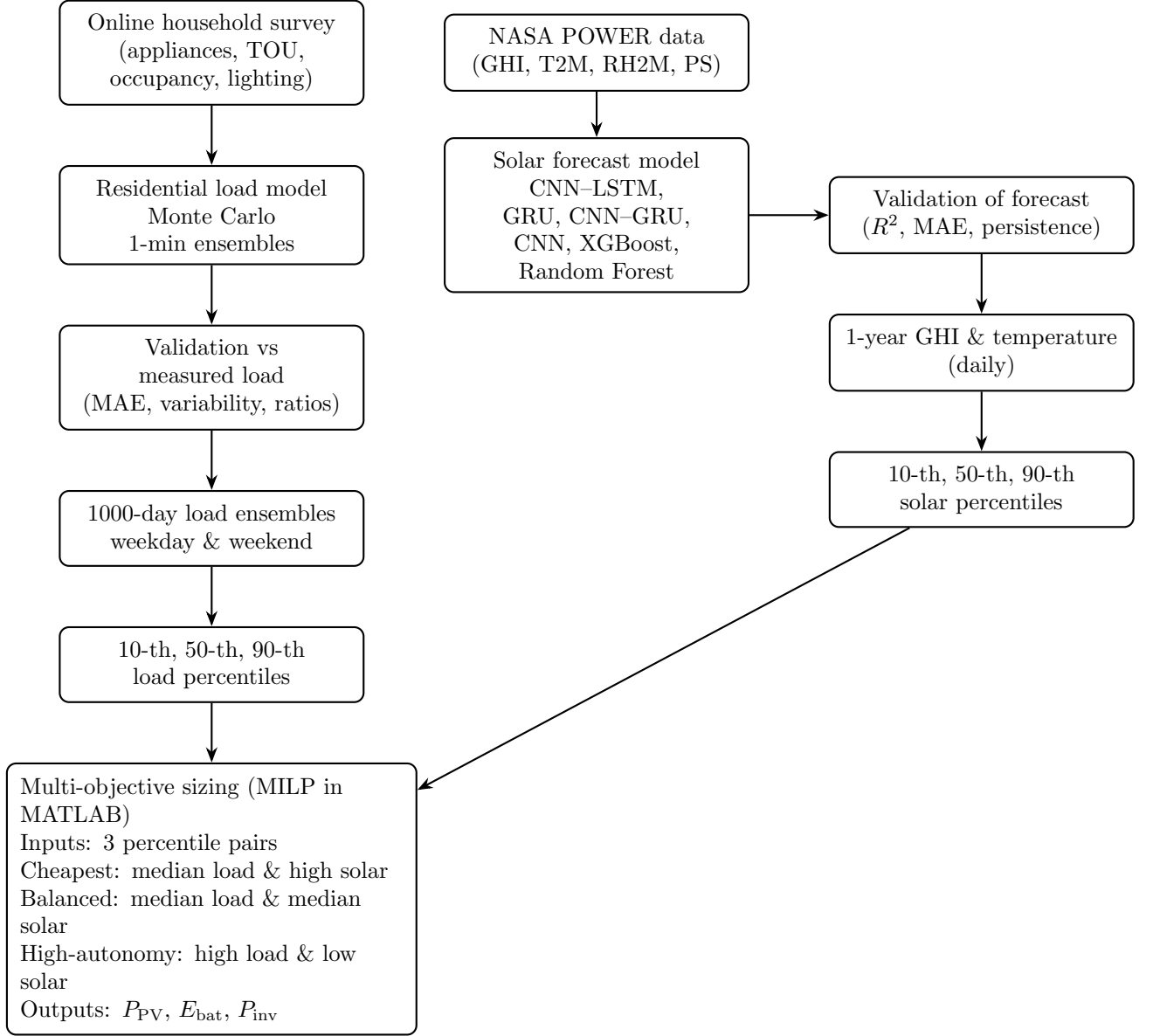


Figure 5: Methodology overview

3.1 Residential Load Modelling

This study develops a stochastic bottom-up household load model explicitly for residential hybrid photovoltaic–battery (PV–battery) system sizing. It adapts the CREST

demand modeling framework [10] with MECS-CREST modifications [12] for household-specific load applications. The methodology has three steps, namely: (1) model parameterisation through an online survey of representative households, (2) Python implementation, and (3) validation with independent case-study measurements on a monitored household.

3.1.1 Data Collection: Online Survey

The online survey focuses on 3 consumption groups: low (<5 kWh/day), medium (5–15 kWh/day), and high (>15 kWh/day), chosen to represent the diversity in residential consumption patterns that solar installers would face. Each residence has a detailed instrument with parameters, which includes number and type of appliances, number and wattage of lighting, usage patterns, TOU probabilities, number of persons in residence by hour (weekday / weekend) and usage patterns.

This granular survey data directly inputs a unique model instance per household, which builds a unique Markov transition matrix from its unique occupancy schedule, a complete list of appliances owned by that household with their characteristics, and a probability of TOU/activity based on day type that reflects the behavioral reality of that household. The methodology avoids the use of population averages, as used in the CREST framework [10], and instead uses accurate measurements for each individual household, such that each created load profile accurately represents the PV and battery usage dynamics of each individual household, and can be used for accurate sizing of PV–battery systems.

3.1.2 Model Development

The model development uses a 1-minute resolution stochastic bottom-up load model written in Python, which is parameterized on a per-household basis using the survey data for that household, and uses that household's individual parameters to produce individual load profiles each time the model runs. The occupancy model for each household is a first order Markov chain, whose transition probabilities $P(n_t|n_{t-1})$ are only inferred from the occupancy patterns that the household defines in the survey, so that the activity patterns at the minute-by-minute level for the household can be simulated. n_t represents the number of active residents at minute t for that particular household. The appliance model then uses the time-of-use (TOU) of each of the surveyed appliances i to compute the switch-on probability $p_{i,t} = f(\text{TOU}_{i,t}, n_t)$ for that particular instance, with the TOU and power demand randomly drawn from the underlying distributions of the appliance model, based on the appliance type, count, rated power, and time-of-use probability for that appliance. The model for the lighting in the household calculates the probability of activation $p_{\text{light},t} = g(\text{irradiance}_t, \text{bulbs}, n_t)$, depending on the number of bulbs in the household, the bulb wattage, the usage rates and the NASA POWER 1-minute clearness index and applies this to modulate the lighting needs as a function of the available natural light and the occupancy dynamics in the household. These Monte Carlo ensembles consist of more than 1000 runs for each of the household instances, generating 24 hours of load profiles at 1-minute intervals. Each ensemble represents a weekday and weekend profile with realistic day-to-day variations in appliance usage and occupancy changes in response to stochastic behavior in the Markov transitions, TOU probabilities, and environmental variables.

3.1.3 Model Validation

The case study for empirical validation is chosen from the surveyed household cohort and comprehensive measurements are made to capture the occupancy pattern, appliance ratings, time-of-use behaviors and the actual 1-minute load data are continuously recorded during 30 days by a high accuracy meter. Two distinct Monte Carlo ensembles are created in this case, one consisting of 1000+ simulation runs over the weekdays, and a second consisting of 1000+ simulation runs over the weekends, using the exact parameters from the survey in the case-study household to create a weekday ensemble and a weekend ensemble respectively. Three ensembles are extracted from each ensemble and are used for model validation: the max usage profile (largest peak usage across all runs), the min usage profile (smallest peak usage) and the average usage profile (average of all runs). This allows for a detailed assessment of the model's ability to represent both extreme and typical load behaviour over the course of a day.

These extracted profiles are compared with the corresponding measured data segmented by day type using quantitative measures such as Mean Absolute Error (MAE) defined as

$$\text{MAE} = \frac{1}{N} \sum_{t=1}^N |L_{\text{model},t} - L_{\text{meas},t}|, \quad (1)$$

variability ratio expressed as $\sigma_{\text{model}}/\sigma_{\text{meas}}$, peak-to-average ratio comparisons for each profile type (max, min, average), and visual inspection of 48-hour composite profiles combining weekday and weekend periods. Validation is performed by constraining the model to meet performance thresholds: MAE must be $\leq 10\%$ for all three profile types; variability ratio $\leq 0.9 - \geq 1.1$; peak/average ratios must be within 10% of measured values, with targeted parameter refinement (sensitivity analysis) of either the household-specific Markov transitions or the appliance TOUs probabilities or lighting irradiance thresholds

in order to ensure the model represents both routine operations and the behaviour of boundary conditions observed in the case-study measurements.

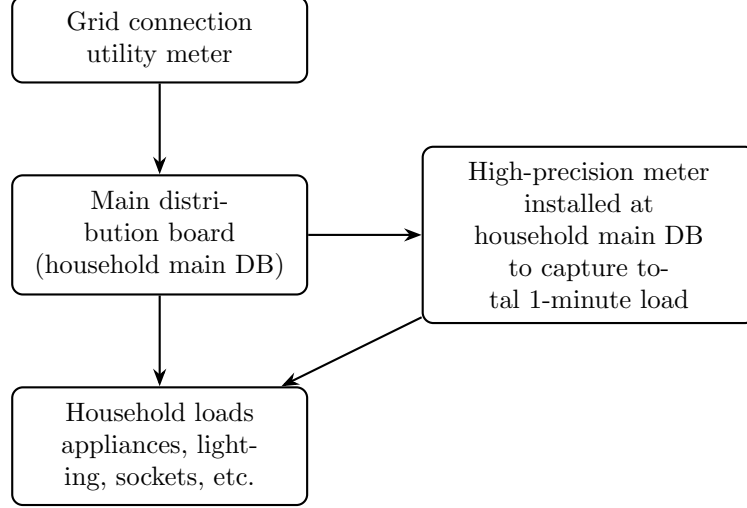


Figure 6: Experimental setup at the case-study household.

3.2 Long-Term Solar Irradiance and Temperature Forecasting

This study develops a physics-informed deep learning ensemble for one-year-ahead Global Horizontal Irradiance (GHI) and temperature forecasting at daily resolution, extending Azizi et al. [13] methodology to Kenyan conditions while incorporating Ghodusinejad et al.’s physics-hybrid recommendations [14]. The methodology uses daily NASA POWER data as the primary learning signal and forecasts at daily resolution for use in long-term PV planning. The six models benchmarked are convolutional neural network–long short-term memory (CNN–LSTM), gated recurrent unit (GRU), convolutional neural network (CNN), CNN–GRU, extreme gradient boosting (XGBoost), and random forest (Random Forest), all enhanced with a Clear Sky Index (CSI) physics feature.

3.2.1 Data acquisition, temporal resolution shift, and feature engineering

The methodology starts from NASA POWER daily time series (1981–2025) at Nairobi (1.2921°S, 36.8219°E) for GHI (`ALLSKY_SFC_SW_DWN`) and 2 m air temperature (T2M). The daily data are quality-controlled, missing values are filled using climatological interpolation, and the series are converted into a supervised learning format using a sliding window over daily values. Baseline inputs comprise daily GHI, T2M, 2 m relative humidity (RH2M), and surface pressure (PS). A physics-informed Clear Sky Index (CSI) feature is added, defined as

$$\text{CSI}_t = \frac{\text{GHI}_{\text{measured},t}}{\text{GHI}_{\text{clear-sky},t}}, \quad (2)$$

where $\text{GHI}_{\text{clear-sky},t}$ is computed via the Ineichen clear-sky model with Linke turbidity $T_L = 3.5$ representative of Nairobi. The daily feature matrix is thus formed as sequences of $60 \text{ days} \times 5 \text{ features}$ (GHI, T2M, RH2M, PS, CSI), yielding 300 scalar inputs per training sample, which is used for all deep learning models. Tree-based models are used with these same 300-dimensional feature vectors, except now they are no longer dynamic inputs, but rather static ones.

3.2.2 Temporal data split and rolling forecast logic

The methodology adopts a strict chronological time split with 1981–2013 used as a training set, 2014–2018 as validation set, and 2019–2025 for test set evaluation. The final 2026 forecasts are created using the training- and validation data as one time series for each model. A rolling forecasting origin is used within the test period: For every day t in the test and forecast horizon the model is presented with the previous 60 days of data (from $t - 59$ to t), and is asked to forecast the next 30 days (from $t + 1$ to $t + 30$). This is a rolling window, which is pushed one day ahead each day, to get a model to look

at the entire one year ahead.

3.2.3 Model formulations and training workflow

The six models are trained on the 60-day input window aggregated to 30-day outputs in daily resolution. CNN-LSTM first applies a 1D convolutional layer (32 filters, kernel size 3) over the 60-day sequence to extract local patterns, then feeds the resulting feature maps into a two-layer long short-term memory (LSTM) network (64 hidden units per layer) that maintains temporal dependencies across months. GRU processes the same 60-day input directly with a two-layer GRU (64 hidden units per layer), using update and reset gates to control memory flow according to

$$\mathbf{h}_t = (1 - z_t) \odot \mathbf{h}_{t-1} + z_t \odot \tilde{\mathbf{h}}_t, \quad (3)$$

with the final hidden state mapped to the 30-day output. CNN-GRU combines the two previous ideas: a 1D convolutional neural network (CNN) with 32 filters and kernel size 3 first produces feature maps

$$\mathbf{F}_t = \text{ReLU}(\text{Conv1D}(\mathbf{X}_{t-59:t}, 32 \text{ filters, kernel} = 3)), \quad (4)$$

which are then passed to a two-layer GRU (64 hidden units per layer) that learns how these patterns evolve over time,

$$\mathbf{h}_t = \text{GRU}(\mathbf{F}_{t-59:t}, \mathbf{h}_{t-1}), \quad (5)$$

before mapping to the 30-day forecast. The pure CNN model uses the same 32-filter convolutional layer to extract patterns but then applies fully connected layers to produce the 30-day output without recurrent components. Finally, extreme gradient boosting (XGBoost) and random forest (Random Forest) are trained on the 300-dimensional feature vectors (60 days $\times$ 5 features) as static inputs, with 500 trees and maximum depth 6

used for both. Hyperparameters are tuned on the validation set, and the best-performing model configuration is retrained on the combined 1981–2025 dataset for the final 2026 forecast.

3.2.4 Evaluation and model selection

All models are evaluated on the test set (2019–2025) using the daily forecasts with persistence (next day equals current day) as a baseline. Metrics used are the coefficient of determination (R^2), the mean absolute error (MAE), and the gain in improvement over persistence for GHI and temperature. For the best-performing deep learning model, uncertainty is quantified by Monte Carlo dropout (100 stochastic forward passes) to construct 80% and 95% prediction intervals at the daily level. The final model is selected based on test-set performance, uncertainty calibration, training stability, and computational efficiency, and then applied to the full 1981–2025 period to generate one-year-ahead daily forecasts for 2026, from which percentile-based daily profiles are extracted for the MILP sizing optimization.

3.3 Multi-Objective Sizing Optimization

This study uses a mixed-integer linear programming (MILP)–based multi-objective optimization in MATLAB to size hybrid PV–battery–grid systems for each household. Representative load and solar profiles are selected from the 1000-day load ensembles and 365-day solar forecast using the 10-th, 50-th, and 90-th percentiles of daily energy and peak power. Three Pareto-optimal designs are generated: cheapest, balanced, and high-autonomy, each optimized for one specific percentile combination, while the full ensembles are saved for post-optimization validation.

3.3.1 Representative Case Selection Using Percentiles

For each household, daily statistics (total energy and peak power) are computed from the 1000-day, 1-minute load profiles, and the 10-th, 50-th (median), and 90-th-percentile days are chosen as low-load, median-load, and high-load representative cases. The 365-day solar forecast is used to define low solar, median solar and high solar representative days by using similar percentiles. This is then applied when the MILP is run with the combinations listed in Table 3:

Table 3: Percentile-based input combinations for each design mode

Design Mode	Load Percentile	Solar Percentile
Cheapest	50-th (median)	90-th (high)
Balanced	50-th (median)	50-th (median)
High Autonomy	90-th (high)	10-th (low)

3.3.2 System Model Formulation

The system consists of a photovoltaic (PV) array, LiFePO₄ battery, grid connection, and hybrid inverter, all optimized for a single 24-hour representative day. PV power is modeled as

$$P_t^{\text{PV}} = P_{\text{PV}} \cdot \frac{G_t}{G_{\text{STC}}} \cdot [1 + \gamma(T_t - T_{\text{STC}})], \quad (6)$$

where P_{PV} is the installed PV capacity, and G_t , T_t are from the selected percentile-based solar profile. The battery state of charge (SOC) is updated as

$$\text{SOC}_{t+1} = \text{SOC}_t + \eta_c P_t^c \Delta t - \frac{P_t^d}{\eta_d} \Delta t \quad (7)$$

with SOC constrained between 0.1 and 0.9 (10%–90% SOC, 10%–80% depth of discharge, DOD) and $\eta_c = 0.95$, $\eta_d = 0.98$. The inverter rating P_{inv} satisfies

$$P_{\text{inv}} \geq \max(P_{\text{PV}}, P_{\text{bat}}). \quad (8)$$

3.3.3 Decision Variables and Objectives

The sizing variables are

$$x = [P_{\text{PV}}, E_{\text{bat}}, P_{\text{inv}}] \quad (\text{kW}, \text{kWh}, \text{kW}), \quad (9)$$

along with hourly operational variables P_t^{PV} , P_t^c , P_t^d , SOC_t , and binary indicators u_t^c , u_t^d for charging and discharging. The optimization minimizes total cost

$$f_1(x) = c_{\text{PV}}P_{\text{PV}} + c_{\text{bat}}E_{\text{bat}} + c_{\text{inv}}P_{\text{inv}} + c_{\text{BOS}} \quad (10)$$

and maximizes self-sufficiency (energy fraction served by PV and battery) via

$$f_2(x) = -\frac{\sum_{t=1}^T E_t^{\text{served}} \cdot w_t}{\sum_{t=1}^T L_t \cdot w_t} \quad (11)$$

with $E_t^{\text{served}} \leq P_t^{\text{PV}} + P_t^d$ and $E_t^{\text{served}} \leq L_t$ to linearize the min operation.

3.3.4 Constraints and Solution Method

Load balance, PV limits, and battery constraints are enforced as

$$L_t = P_t^{\text{PV}} + P_t^d + P_t^{\text{grid}} + E_t^{\text{ns}}, \quad P_t^{\text{PV}} \leq P_{\text{PV}} \cdot \frac{G_t}{G_{\text{STC}}}, \quad (12)$$

$$P_t^c \leq P_{\text{bat}}u_t^c, \quad P_t^d \leq P_{\text{bat}}u_t^d, \quad u_t^c + u_t^d \leq 1, \quad 0.1 \leq \text{SOC}_t \leq 0.9. \quad (13)$$

For high-autonomy, a days-of-autonomy constraint is added:

$$E_{\text{bat}} \geq D \cdot \bar{L}_{\text{daily}} \cdot \frac{1}{\eta_d \cdot \text{DOD}} \quad (14)$$

with $D = 1$ day, $\eta_d = 0.98$, $\text{DOD} = 0.7$. The ε -constraint method is used in MATLAB for each design mode:

$$\min f_1(x) \quad \text{subject to} \quad f_2(x) \geq \epsilon_i, \quad i = 1, \dots, N_\epsilon \quad (15)$$

yielding three Pareto fronts. From each, the cheapest, balanced, and high-autonomy designs are identified as the minimum-cost, knee, and maximum-self-sufficiency points, respectively, using the percentile combinations in Table 3.

4 EXPECTED RESULTS

The following results are expected at the end of this research:

1. A validated probabilistic load model capable of generating realistic high-resolution synthetic demand profiles for the case-study household.
2. A tested machine-learning model that provides accurate long-term GHI forecasts and corresponding PV energy yield time series, including basic uncertainty estimates for the study location.
3. A validated MILP-based sizing method that generates Pareto-optimal PV–battery–grid system sizes, summarized into three practical design options (cheapest, balanced, high-autonomy) with quantified objective indicators.

5 BUDGET

Table 4 shows the project budget breakdown.

Table 4: Budget Breakdown		
Item	Description	Cost (KES)
Laptop	Mid-range laptop for simulations and ML models	80,000
Power meter	Single-phase DIN-rail meter for household load monitoring	40,000
Transport	Matatu/taxi fares for field visits	3,000
Misc. expenses	Printing, data bundles, stationery	7,000
Total		130,000

6 WORK SCHEDULE

The research is expected to take approximately twelve (12) months. This time is distributed among the various activities as shown in Figure 7.

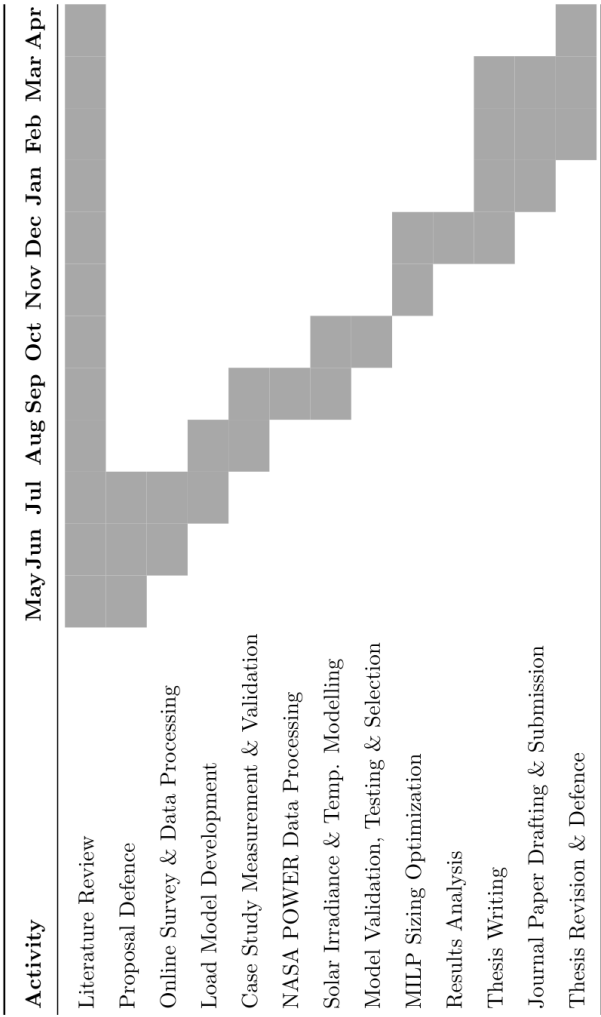


Figure 7: Work schedule for the research project.

References

- [1] International Energy Agency, “World energy outlook 2023,” tech. rep., IEA, 2023.
- [2] International Energy Agency, “Renewables 2024: Analysis and forecast to 2030,” tech. rep., IEA, 2024.
- [3] J. C. León Gómez, S. E. De León Aldaco, and J. Aguayo Alquicira, “A review of hybrid renewable energy systems: architectures, battery systems, and optimization techniques,” *Eng*, vol. 4, no. 2, pp. 1446–1467, 2023.
- [4] S. Ahmed, A. Ali, J. A. Ansari, S. A. Qadir, and L. Kumar, “A comprehensive review of solar photovoltaic systems: scope, technologies, applications, progress, challenges and recommendations,” *IEEE Access*, 2025.
- [5] K. E. Fahim, “Off grid solar pv system sizing for a typical east african household,” 2023.
- [6] B. Pirouz, F. Guerriero, and A. Algieri, “Optimal sizing and operation of battery energy storage systems in renewable energy systems using multi-objective optimization and artificial intelligence,” 2024. Preprint available at SSRN 5442697.
- [7] B. Aboagye, S. Gyamfi, E. A. Ofori, and S. Djordjevic, “Investigation into the impacts of design, installation, operation and maintenance issues on performance and degradation of installed solar photovoltaic (pv) systems,” *Energy for Sustainable Development*, vol. 66, pp. 165–176, 2022.
- [8] R. Nematirad, A. Pahwa, B. Natarajan, and H. Wu, “Optimal sizing of photovoltaic-battery system for peak demand reduction using statistical models,” *Frontiers in Energy Research*, vol. 11, p. 1297356, 2023.

- [9] X. Chen, Z. Liu, P. Wang, B. Li, R. Liu, L. Zhang, and S. Luo, “Multi-objective optimization of battery capacity of grid-connected pv-bess system in hybrid building energy sharing community considering time-of-use tariff,” *Applied Energy*, vol. 350, p. 121727, 2023.
- [10] E. McKenna and M. Thomson, “High-resolution stochastic integrated thermal-electrical domestic demand model,” *Applied Energy*, vol. 165, pp. 445–461, 2016.
- [11] P. T. Kapen, “Multi-objective optimization of a wind/photovoltaic/battery hybrid system using a novel hybrid meta-heuristic algorithm,” *Energy Conversion and Management*, vol. 327, p. 119533, 2025.
- [12] M. Leach, “Household load modelling working paper,” tech. rep., University of Loughborough, jul 2020. Working Paper 30/07/20.
- [13] M. Azizi, M. Yaghoubirad, M. Farajollahi, and A. Ahmadi, “Deep learning-based long-term global solar irradiance and temperature forecasting using time series with multi-step multivariate output,” *Renewable Energy*, vol. 206, pp. 135–147, 2023.
- [14] M. H. Ghodusinejad, N. Rashvand, F. Salmanpour, S. Danehkar, and H. Yousefi, “A systematic review of solar irradiance forecasting across time horizons using physical, satellite, and ai-based methods,” *Solar Compass*, p. 100154, 2025.
- [15] N. Promasa, E. Songkoh, S. Phonkaphon, K. Sirichunchuen, C. Ketkaew, and P. Unahalekhaka, “Optimization of sizing of battery energy storage system for residential households by load forecasting with artificial intelligence (ai): Case of ev charging installation,” *Energies*, vol. 18, no. 5, p. 1245, 2025.

- [16] J. P. Fossati, A. Galarza, A. Martín-Villate, and L. Fontan, “A method for optimal sizing energy storage systems for microgrids,” *Renewable Energy*, vol. 77, pp. 539–549, 2015.
- [17] C. Lu, H. Xu, X. Pan, and J. Song, “Optimal sizing and control of battery energy storage system for peak load shaving,” *Energies*, vol. 7, pp. 8396–8410, 2014.